Report on the paper "A Branch-and-cut and MIP-Based Heuristics for the Prize-Collecting Travelling Salesman Problem"

September 2019

1 General comments

The paper presents a Branch-and-cut algorithm (BC) and math-heuristics for the Prize Collecting Traveling Salesman Problem (PCTSP). The authors compared the BC with Branch-and-bound algorithms based on other formulations from the literature and their math-heuristics with other heuristic approaches published elsewhere.

The main contribution of the paper is the introduction of the Relax-and-fix and Diving heuristics for the problem. The Branch-and-cut algorithm is quite basic. I do not think it is a real contribution, despite the fact that it outperforms other algorithms based on different formulations (network flow models). The formulation behind the algorithm is pretty standard for the TSP and its variants. Better computational results seem to emerge much more from the fact that a BC based on the cutset formulation had never been implemented for that particular problem that from its own merits. In that respect, much more could have been done. I understand that the merit of paper is to use a simple PCTSP integer programming formulation (and BC based on it) to drive the heuristics.

My concern about the paper has to do with the poor writing. There are a lot of grammar mistakes and typos. Even the mathematical formulation is wrongly stated. It gives me the impression that the paper was written in a lazy fashion. Thus, I think the paper cannot be accepted as it is. I suggest a major revision to address these issues.

2 Technical questions

1. Concerning Algorithm 1: 1) How ties are dealt with when choosing the variable with the least fractionality ? 2) Why not implementing back-tracking in RF when the resulting model is infeasible ?

2. For the general description of the Relax-and-fix, the authors state that after fixing set Q_n, for $n = 1, \ldots, P$, the *model is solved.* But what is solved is a continuous relaxation, where some variables were fixed as some integer values, some are constrained to be integers and others are free. But for the particular PCTSP case, only variables $y_i : i \in V$ are candidates to be included in the sets defining the partition. This is what I understood from the last phrase of the first paragraph of p. 5. How are the partitions defined in that case ? The writing does not help. In summary, the Relax-and-fix needs to be better explained. I suggest being very precise: indicate the mathematical formulation of the problem being solved after each round of fixing variables. Please, indicate in such a model which are the sets of variables that are integer constrained

but free, integer and fixed and relaxed. That should replace the general and unclear statement $x^* \leftarrow solve(model)$ used in the description of the algorithm.

3. The comparison of computational results between the math-heuristics and CS* does not seem fair to me. Please, correct me if I am wrong.
   As far as I understood, computational results from CS* were carried over from the literature, since its source code is not available and CS* was not re-run. Hence, that means that the CPU times of CS* and the RF and diving heuristics are not directly comparable. Only the quality of the upper bounds should be compared. Please, re-phrase that last statement of the last paragraph of the computational results or, give clear indication that the proposed heuristics are in fact faster than CS*.

3 Grammar mistakes, typos, etc.

In case of revising the manuscript, please, number the lines in each page. Thank you.

1. Abstract:
   (a) This paper approaches, this paper suggests.
   (b) ...were conducted in instacnes, with instances.
   (c) ...showed the efficiency of our methods.
   I believe *efficiency* is a reserved word in optimization, that ultimately means *polynomially solvable*. Since the algorithms do not run in polynomial time, I think this phrase could be re-written.

2. Section 1:
   (a) p.1, first paragraph: consists of to determine, consist of determining.
   (b) p.2, first line: if the vertex is left unrouted. Please add a comma after unrouted.
   (c) p.2, ...while collecting a minimum prize p_{min} is guaranteed by a knapsack constraint. I think there is no need to state that the knapsack constraint is going to be enforced, as long as, for the moment, PCTSP is being defined. The phrase could end right after prize p_{min}.
   (d) p.2, an example PCSTP solution, a PCTSP solution.
   (e) p.2: [12] presented some bounding... I think Fischetti and Toth [12] is a better way of starting the same phrase. Please correct other phrases in the paper where the number of the reference actually makes the role of the subject of the phrase.
   (f) p.2: ...new formulation for PCTSP, in which uses cut-set. In + which does not go along together here; please remove the word in and the comma, so the phrase becomes: new formulation for PCTSP which uses cut-set...
   (g) p.2 (same paragraph): The author also developed. The authors.
   (h) p.2: presented a Lagrangian heuristic to obtain an upper bound in the form of a solution. It is not clear to me what the purpose of the underlined words in the phrase. Wouldn't it be simpler to state just: an upper bound ?
   (i) p.2: ,by relaxing the minimum premium constraint and applying the sub-gradient method. I think that, to make sense, it needs to be stated that besides being relaxed, the inequality was also dualized in a Lagrangian way, so the subgradient method was used.

(j) p.2: The Quota-TSP was not defined.

(k) p.2: ii) two heuristics a based on clustering search...This phrase is weird, please correct. Is would it be two heuristics based on clustering...?

(l) p.2, in which the authors propose, proposed.

3. Section 2: Mathematical formulation.

   (a) edge $e = ij$. Later on, edge is referred to as (i, j). Please choose one form and stick to it.

   (b) Equality (4) must be an inequality in the $\geq 2y_i$ form.

   (c) Constraints (3) imposes. Impose.

   (d) (5) and (6) impose that all variables be 0-1. are 0-1, or need to be 0-1.

4. Section 3: Branch-and-cut algorithm.

   (a) p. 3: However, the violation of such constraints can be checked in polynomial time, making a branch-and-cut algorithm interesting.
   I understand what the authors meant, but I have some concerns about this phrase. First, the meaning of interesting is rather vague. Secondly, even if the separation problem were not polynomially solvable, a branch-and-cut based on them could still be effective, provided that 1) these inequalities could be separated efficiently for integer points and by that I mean polynomially separated and 2) could be separated heuristically for fractional points, in order to strengthen the LP bounds. So, overall I believe this phrase needs improvements.

   (b) p. 3: As usual, the family of exponential size constraints (5). ...constraints (4)

   (c) p. 4: are integrated into the current formulation and the strengthened formulation is solved. .... to the current relaxation , and the strengthened relaxation is solved (not formulation in both pieces).

   (d) p. 4: We used the Mixed Integer Linear Programming (MILP) Gurobi.
   To what purpose ? To solve the LP relaxations, to implement the search tree ? I think that both apply.

   (e) p. 4: In the initial formulation, we include all constraints (2),(3).... in the initial relaxation (again, it is not a formulation).

   (f) p. 4: leaving out the cutset constraints (5)... cutset constraints (4).

   (g) p. 4: ..$S = (x^*, y^*)$ obtained though. let $S = (x^*, y^*)$ be an optimal solution to the current relaxation....

   (h) p. 4: when the LP solution S is integer ou fractional. or fractional.

   (i) p. 4: if the output is less than 2. less than $2y_i$.

   (j) p. 4: Goldberg's pre-flow algorithm. pre-flow push algorithm.

5. Section 4: MIP Heuristics

   (a) p. 4: Heuristic methods are often used, since they can produce viable solutions as quickly as possible. This phrase is far too general. Does it always apply ? I am afraid it does not. I believe it would be better stating like this: since, in several cases, they provide feasible solutions quickly....

   (b) p. 4: the number of sets P.

(c) In an iteration n, only the Q_n set variables. For each iteration n, only the set of variables in Q_n.

(d) p.5: fixed at the current values. at their current values.

(e) p.5: are fixed at their current value, at their current values.

(f) p. 5: and the process repeats for the remain sets. is repeated for the remaining sets.

(g) p. 5: the most state-of-the-art MIP solvers. the state-of-the art, remove the most.

(h) p. 5: please decide whether or not to capitalize Branch-and-bound throughout the text or not.

(i) p. 5: to find optimal solutions. on the other hand,... On the other hand.

(j) p. 5: in order to find ... quickly, is desirable. it is desirable.

(k) p. 5: on bounding variable. on bounding variables.

(l) p. 6: Thus, we our approach. Please choose one.

(m) p. 6: one selects the a fractional variable. Again, choose one. The indefinite form is the appropriate to choose.

6. Section 5:

   (a) p. 6: the instances consists. consist.

7. References. Please, revise the entire section. There are some incomplete references, for instance, [5,6,8,16] and [19] whose publication date is wrong.